PicoMagnetometer RPi software User manual

Pico^{nT}

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UK Radio Astronomy Association

Python code for the UKRAA PicoMagnetometer

Set of Python, gnuplot and shell scripts to run on an RPi4/5 to record, process and present data from the UKRAA PicoMagnetometer.

This software was written to suit a specific set-up; feel free to use as you see fit.

Instructions for initial setting up of a Raspberry Pi4/5 are included in the **docs** folder on the supplied microSD card for your UKRAA PicoMagnetometer

Table of Contents

Using the code

The code assumes username is **pi**.

If **pi** is not the username, then you will need to change all occurrences of '**/home/pi**' to '**/home/*username***' in the python, gnuplot and shell scripts, where ***username*** is the username you have selected for your RPi4/5.

The code assumes that your UKRAA PicoMagnetometer is connected to the RPi4/5 USB and that it will be connected via **/dev/ttyACM0.**

If there are other devices connected to the RPi and your magnetometer is not **/dev/ttyACM0**, then you will need to change **/dev/ttyACM0** to **/dev/*ttyACMx*** in the **GetDataRawACM0.py** python script, where ***ttyACMx*** is the tty address of you connected magnetometer.

GetDataRawACM0.py is run as a service.

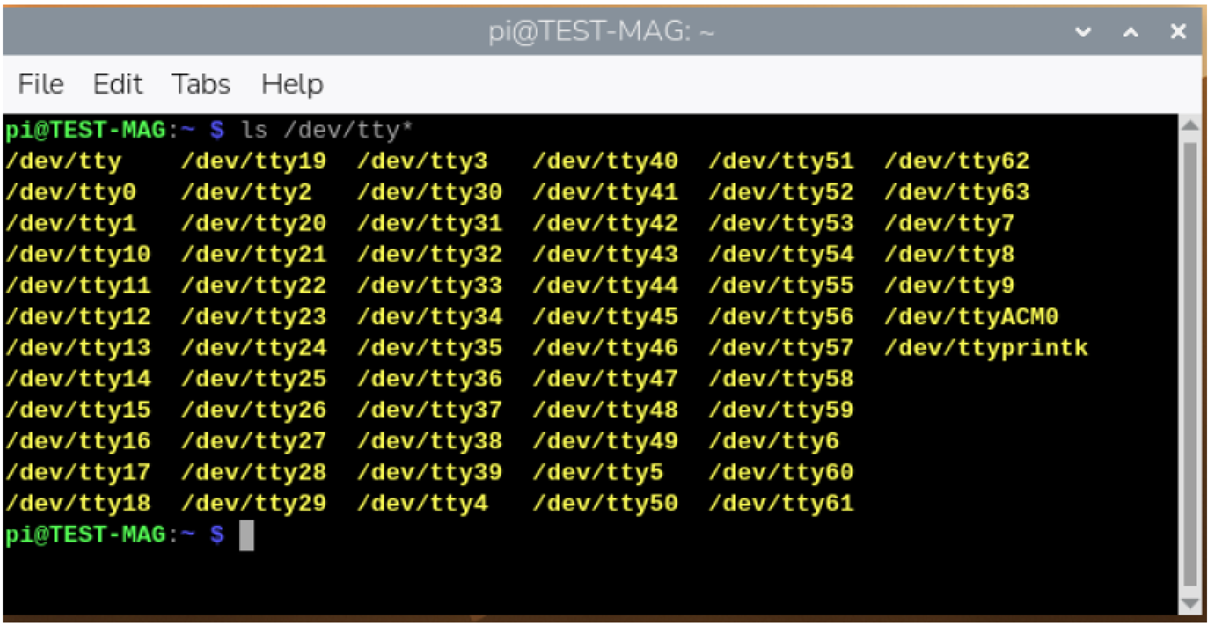
Other scripts (Python and gnuplot) are run from **cron** using shell scripts.

Where is my PicoMagnetometer?

Plug your PicoMagnetometer into any of the RPi USB ports - I normally use one of the blue ports (USB3).

1. Open a terminal window and type the following command and press enter

ls /dev/tty*



2. You are looking for **/dev/ttyACM0** - this is on the right hand side of the screen shot above.

3. This is the USB address for your attached PicoMagnetometer - if you have more than one magnetometer attached you may see **/dev/ttyACM1** etc.

4. If you do not see **/dev/ttyACM0**, then unplug and plug the PicoMagnetometer back in and try again.

5. As long as you see **/dev/ttyACM0** then you do not have to make any changes to the python scripts, because they are looking for **ACM0**.

Getting the software onto your RPi

1. Log into your Raspberry Pi4/5 using VNC.

2. Open a terminal window, type the following command and press enter

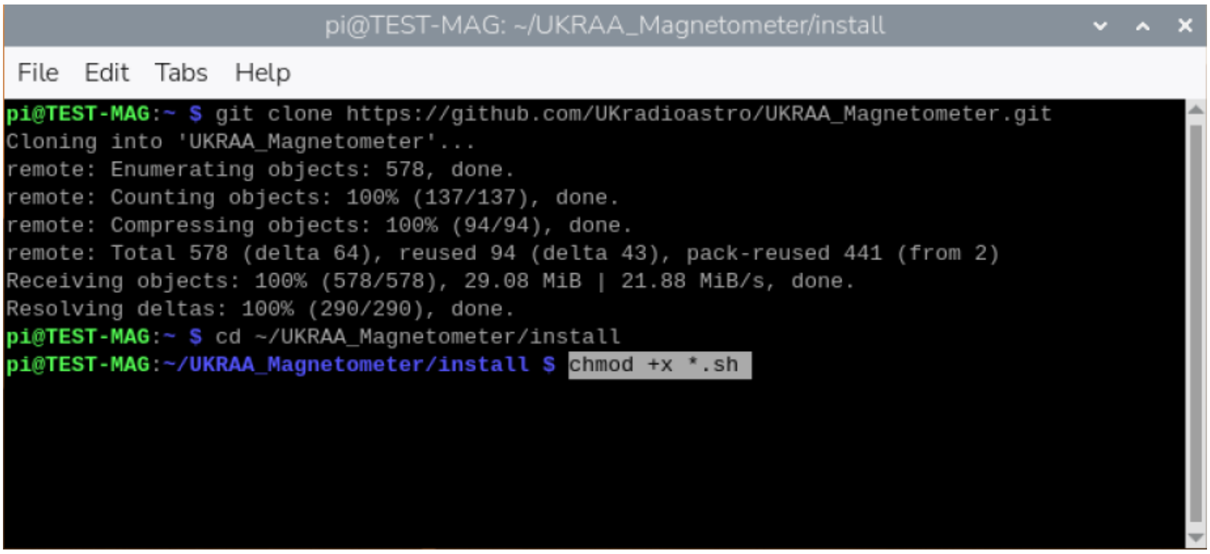
   git clone https://github.com/UKradioastro/UKRAA_Magnetometer.git

This will download all of the code to the directory **UKRAA_Magnetometer** inside **/home/pi**

Installing the software onto your RPi

1. Open a terminal window and type the following command and press **enter**

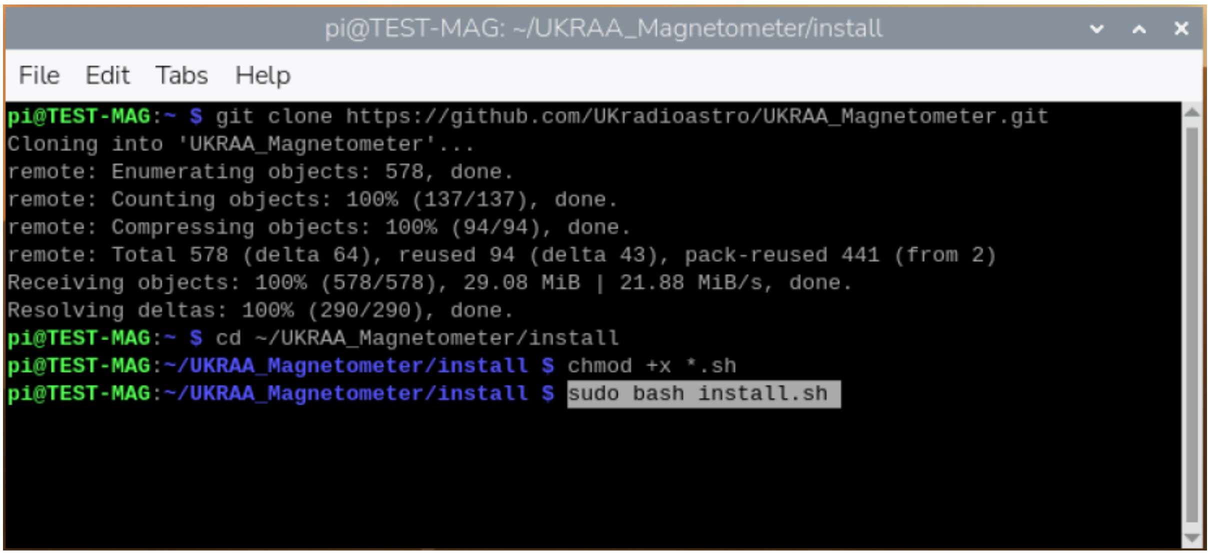
   chmod +x ~/UKRAA_Magnetometer/install/*.sh



This will make the **install.sh** script executable.

2. Type the following command and press **enter**

   sudo bash ~/UKRAA_Magnetometer/install/install.sh



This will run the install script.

There may be occasions during the running of the install script that require you to make a keyboard entry.

When asked **Do you want to continue? [Y/n]** - type **Y** or **y** and press **enter**

That's it!

The code is now set up to run automatically; it will record the data from the PicoMagnetometer, process the data, plot the data and post the plots to your intranet web page.

There are other functions that are customisable by the user, such as **email alerts** and **FTP** to users' external website, that are covered in other sections in this manual.

Optional: run an install-time heartbeat smoke check (disabled by default):

sudo MAGNETOMETER_INSTALL_SMOKE_HEARTBEAT=1 bash install.sh

This runs **--test-heartbeat** once during install and prints a clear **HEARTBEAT_SMOKE_CHECK: PASS** or **HEARTBEAT_SMOKE_CHECK: FAIL** summary.

What does the code do?

Things it does right from the start

The code records data from the UKRAA PicoMagnetometer via serial over the supplied USB cable and stores the data to the raw data folder.

The raw data will be processed daily, via CRON, to get magnetic field values per minute in the X (N->S), Y (E->W) and Z (U->D) directions from your previous day's data. The maximum hourly variation of the magnetic field values in the X (N->S) and Y (E->W) directions will also be produced.

A number of plots will be created:

- X, Y and Z (North/South, East/West and up/down)
- Maximum hourly variation in X and Y directions

The raw data will also be processed on a continuous 5 minute basis, again via CRON, to generate a rolling 24 hour plot of X, Y and Z magnetic fields and % change of magnetic field for combined X and Y directions. The latter used to predict the potential of visible Aurora activity.

A simple web server and web page is set up on your RPi so that you can view your PicoMagnetometer results on your desktop PC and/or smart phone when connected to your home network. To access the webpage, see ***PicoMagnetometer webpage*** section for details.

Things it can do with selectable options

Additional plots

There is the option of the following additional rolling and daily plots

- H, D and Z (Local horizontal plane, declination angle and up/down)
- B and I (Total strength of Earth's magnetic field and angle of Earth's magnetic field)

These are configurable through **plot.ini** file, see ***Optional HDZ and/or BI plots*** section for details.

Email alerts

There is the option of receiving email alerts when an activity threshold is passed.

This is configurable through the **alerts.ini** file, see ***Optional Rolling alert Emails*** section for details.

Upload, via FTP, to external hosted website

There is the option of uploading all generated plot to an externally hosted webpage.

This is configurable through the **remote-upload.ini** file, see ***Optional Remote FTP upload*** section for details.

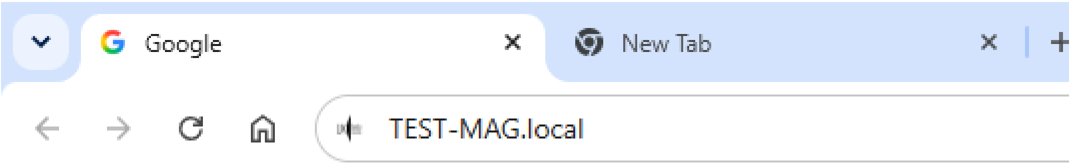
PicoMagnetometer webpage

To access the PicoMagnetomer webpage from your desktop PC or your smart phone…

1. Open your preferred web application (Safari, Chrome, Firefox, etc.).

2. In the search bar type the following and press enter

   http://test-mag.local



This will take you to the web page for your PicoMagnetometer on your local intranet, displaying both the rolling 24 hour plots and yesterday's plots.

NOTE: if you have a different **hostname** for your RPi, change the search bar entry to…

http://*hostname*.local

Where ***hostname*** is the hostname for your RPi setup.

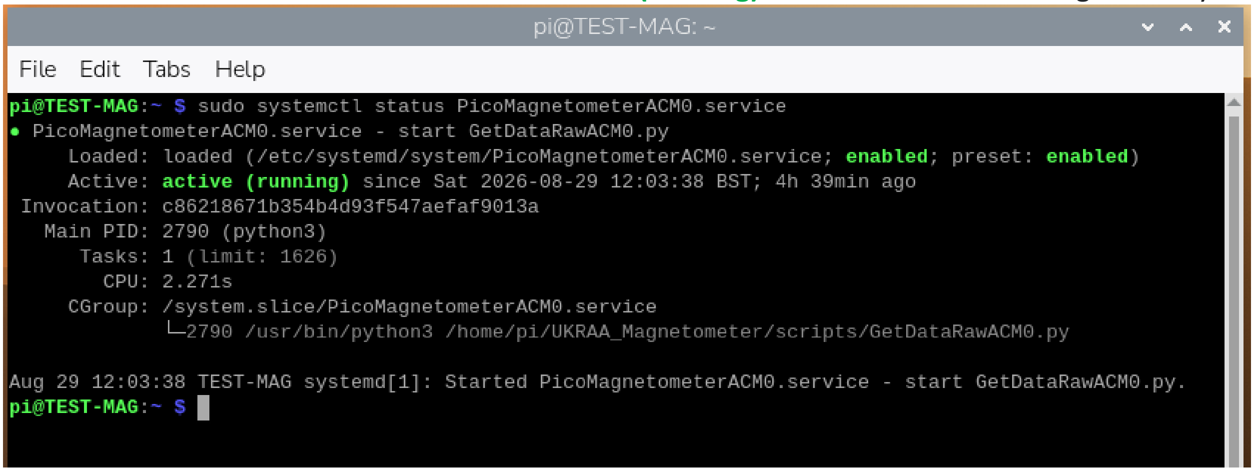
Code is also supplied that will enable the user to upload, via FTP, to their host website, should they have one - see **Optional remote FTP upload** section for more details.

Check PicoMagnetometerACM0 service is running

1. To check the **status** of your service, type the following command and press enter.

   sudo systemctl status PicoMagnetometerACM0.service

   Two **GREEN enabled** and a **GREEN active (running)** indicates service running correctly.



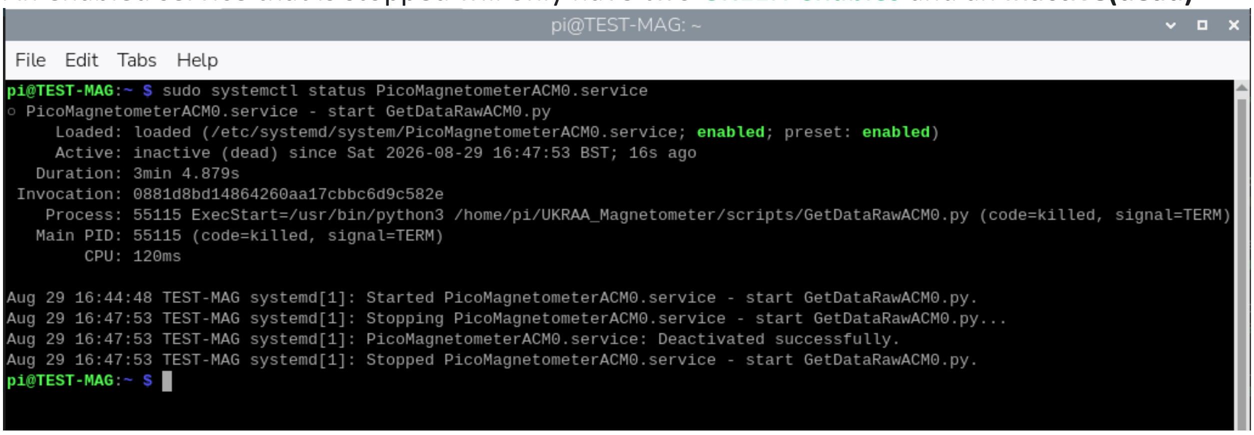
2. To **start** your service, type the following command and press enter.

   sudo systemctl start PicoMagnetometerACM0.service

3. To **stop** your service, type the following command and press enter.

   sudo systemctl stop PicoMagnetometerACM0.service

An enabled service that is stopped will only have two **GREEN enables** and an **inactive(dead)**



If this is the case you will need to:

- Start the service

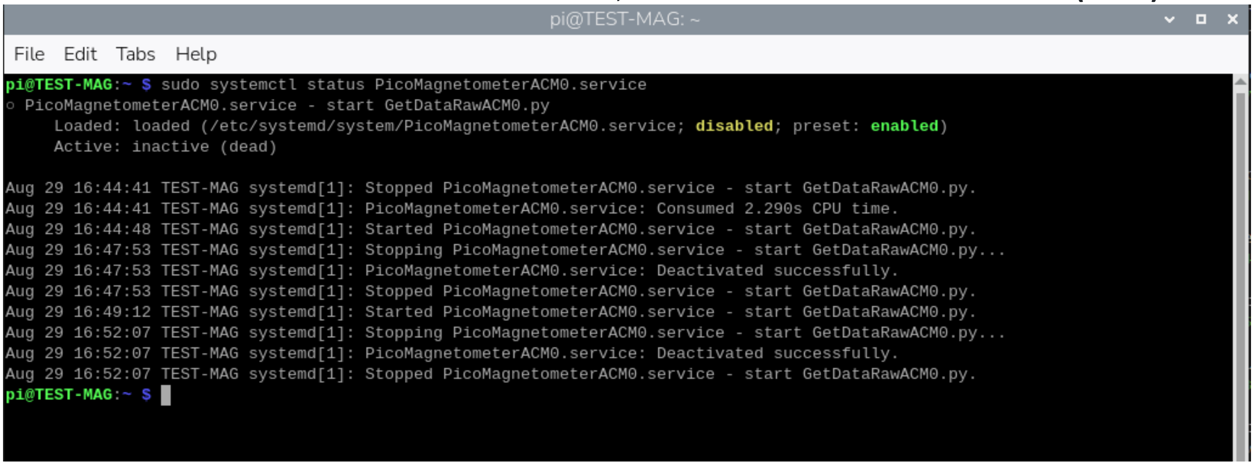
4. To **enable** your service, type the following command and press enter.

sudo systemctl enable PicoMagnetometerACM0.service

5. To **disable** your service, type the following command and press enter.

   sudo systemctl disable PicoMagnetometerACM0.service

A disabled service will have a **YELLOW disabled**, a **GREEN enabled** and an **inactive (dead)**



If this is the case you will need to:

- Enable the service
- Start the service

Optional Post install operational checks

runPostUpdateChecksACM0.sh

After installing/updating scripts on your RPi, you can run:

sudo bash /home/pi/UKRAA_Magnetometer/scripts/runPostUpdateChecksACM0.sh

This executes:

- optional daily publish logic checks
- optional HDZ/BI web visibility checks

checkDailyPublishHealthACM0.sh

For daily runtime monitoring, a health marker is written by:

sudo bash /home/pi/UKRAA_Magnetometer/scripts/checkDailyPublishHealthACM0.sh

Marker output file:

/home/pi/UKRAA_Magnetometer/data/status/daily-health.txt

The marker reports one of three states:

- **PASS** – yesterday was processed and all expected plots are published.
- **PENDING** – no raw data was recorded for yesterday, so there was nothing to process. Expected on the first day after a fresh install, or if the acquisition service was stopped all day. Exit code 0.
- **FAIL** – raw data existed but the processed file or one or more published plots are missing. This is a real fault. Exit code 1.

showDashboardSummaryACM0.sh

One-line dashboard summary on demand:

sudo bash /home/pi/UKRAA_Magnetometer/scripts/showDashboardSummaryACM0.sh

If scheduled in cron, summary snapshots can be collected in:

/home/pi/UKRAA_Magnetometer/logfiles/dashboard-summary.log

Optional HDZ and/or BI plots

Within the **/home/pi/UKRAA_Magnetometer/config** folder there is a file named **plot.ini**.

By default the option of producing HDZ and/or BI plots is turned off (**false**), and the plots do not appear as an option from the intranet webpage.

Should the user wish to have access to HDZ and/or BI rolling plots and yesterday's plots for these options then change the following in the plots.ini file:

- For HDZ plots change line 13
  - From **plot_hdz = false**
  - To **plot_hdz = true**

- For BI plots change line 19
  - From **plot_bi = false**
  - To **plot_bi = true**

These plots will automatically be added to the intranet webpage if selected; with the rolling plot being present after 5 minutes and yesterdays plot being present after 09:30 the next day.

Optional NOAA aurora forecast panel

The webpage can show the latest NOAA Space Weather Prediction Center 30-minute aurora forecast image below the rolling magnetometer status, and this can now be enabled or disabled per installation.

Within the **/home/pi/UKRAA_Magnetometer/config** folder there is a file named **plot.ini**.

By default the option of producing NOAA aurora forecast images is turned off (**false**), and the images do not appear as an option from the intranet webpage.

Should the user wish to have the NOAA aurora forcast image option selected then change the following in the plots.ini file:

[plots]
plot_noaa = false
noaa_hemisphere = north

- **plot_noaa = true enables the panel.**
- **plot_noaa = false disables the panel and removes any stale NOAA image from the published web output.**
- **noaa_hemisphere = north selects the northern aurora forecast.**
- **noaa_hemisphere = south selects the southern aurora forecast.**

The image is downloaded from the chosen NOAA forecast URL, for example:

https://services.swpc.noaa.gov/images/animations/ovation/north/latest.jpg

or

https://services.swpc.noaa.gov/images/animations/ovation/south/latest.jpg

and cached locally as:

/home/pi/UKRAA_Magnetometer/temp/noaa/latest.jpg

The forecast image is updated every 30 minutes by ***updateNoaaAuroraForecast.sh***. If the NOAA download fails, the magnetometer data processing, plotting, alerts and web publishing continue to run.

The webpage includes a source citation and link to the official NOAA product page:

https://www.spaceweather.gov/products/aurora-30-minute-forecast

Optional Rolling alert Emails

Within the **/home/pi/UKRAA_Magnetometer/config** folder there is a file named **alert.ini**.

By default the option of producing rolling email alerts is turned off (**false**).

Should the user wish to receive rolling email alerts from their PicoMagnetometer follow the instructions below;

Rolling alert evaluation supports user defined activity thresholds.

Default values will be:

- **Yellow** at **50nT**
- **Amber** at **100nT**
- **Red** at **200nT**

The user can define their own alert values by modifying lines 18, 19 and 20 of **alert.ini.**

The alert evaluator runs as part of **processRollingData.sh** and only sends email on threshold transitions to levels you choose.

The same threshold values are also used by:

- Daily hourly Activity plot (**PlotDataActivityACM0.gp**)
- Rolling Activity plot (**PlotRollingActivityACM0.gp**)

Configure via alerts.ini file (recommended)

1. Copy the example file: **install/alerts.ini.example**

2. Place it on the Pi as: **/home/pi/UKRAA_Magnetometer/config/alerts.ini**

3. Edit the values for your SMTP service and recipients.
   DO THIS BEFORE CHANGING THE STATE OF email_enabled

4. Set activity thresholds in the same file under **[alerts]**:

   [alerts]
   yellow_threshold_nt = 50
   amber_threshold_nt = 100
   red_threshold_nt = 200

Enable or disable alert emails

Email sending is off by default so a fresh install does not log SMTP failures every 5 minutes.

In **alerts.ini**:

> **[alerts]**
> **email_enabled = true**

Notes:

- When **email_enabled = false**, thresholds, alert levels, plots and the web status page all still work; only email sending is skipped.
- Suppression is logged once per level transition, not on every run.
- The daily heartbeat email is also skipped while **email_enabled = false**.
- The manual test scripts (**testAlertEmailACM0.sh**, **testHeartbeatEmailACM0.sh**) still send, so you can verify SMTP before enabling.
- Environment variable override: **MAGNETOMETER_EMAIL_ENABLED**

By default, **EvaluateAlertsACM0.py** reads: **/home/pi/UKRAA_Magnetometer/config/alerts.ini**

You can override the config path with: **MAGNETOMETER_ALERTS_INI_PATH=/path/to/alerts.ini**

Configure which levels trigger email

Set **MAGNETOMETER_EMAIL_ALERT_LEVELS** as a comma-separated list:

- **RED**
- **RED,AMBER**
- **RED,AMBER,YELLOW**

For instance:

- if you only require **RED** trigger alerts set **email_alert_levels = RED**
- if you require both **RED** and **AMBER** trigger alerts set **email_alert_levels = RED,AMBER**
- if you require all alerts set **email_alert_levels = RED,AMBER,YELLOW**

Configure SMTP and recipients

You can set SMTP/email values in the **alerts.ini** file, or by environment variables. Environment variables take precedence if both are set.

In **alerts.ini:**

> **[smtp]**
> **host = smtp.example.com**
> **port = 587**
> **username = your-smtp-username**
> **password = your-smtp-password**
> **starttls = true**
> **ssl = false**
>
> **[email]**
> **from = magnetometer@example.com**
> **to = you@example.com**
> **attach_plot = false**

Notes:

- If **RollingActivity.png** exists, it is attached to the alert email if **attach_plot = true**

Available environment variables are:

- **MAGNETOMETER_SMTP_HOST** (required)
- **MAGNETOMETER_SMTP_PORT** (default 587)
- **MAGNETOMETER_SMTP_USERNAME** (optional)
- **MAGNETOMETER_SMTP_PASSWORD** (optional)
- **MAGNETOMETER_SMTP_STARTTLS** (***true*** by default)
- **MAGNETOMETER_SMTP_SSL** (***false*** by default)
- **MAGNETOMETER_EMAIL_FROM** (required)
- **MAGNETOMETER_EMAIL_TO** (required, comma-separated for multiple recipients)
- **MAGNETOMETER_EMAIL_ATTACH_PLOT** (***true*** by default)
- **MAGNETOMETER_WEB_URL** (optional, included in email body)

Notes:

- Alert state is stored in **data/alerts/alert-state.json**
- Rolling status is read from **data/status/current.json**
- Alert evaluator config precedence is: environment variable -> **.ini** file -> built-in default

Regenerate a chosen Activity date (quick local test mode)

You can regenerate the hourly Activity plot for any date without storing an archive copy by default:

/bin/bash /home/pi/UKRAA_Magnetometer/scripts/testActivityPlotACM0.sh YYYY-MM-DD

This updates only:

- **/home/pi/UKRAA_Magnetometer/temp/Activity.png**

To also write the dated archive file in **plots/Activity/**, add **--archive**:

/bin/bash /home/pi/UKRAA_Magnetometer/scripts/testActivityPlotACM0.sh YYYY-MM-DD --archive

Send a one-off SMTP test email

To verify SMTP settings without waiting for a threshold transition, run:

/bin/bash /home/pi/UKRAA_Magnetometer/scripts/testAlertEmailACM0.sh

Or run the Python script directly:

/usr/bin/python3 /home/pi/UKRAA_Magnetometer/scripts/EvaluateAlertsACM0.py --test-email

The test email does not update transition state, so normal alert logic is unaffected.

Optional daily heartbeat email

You can enable a once-per-day summary email so you know the system is alive even when no threshold transition occurs.

In **alerts.ini:**

```
[heartbeat]
enabled = true
hour_utc = 9
to =
attach_plot = false
```

Notes:

- **hour_utc** is the first UTC hour of the day when the heartbeat can send.
- Only one heartbeat attempt is made per UTC day (success or failure), to avoid retry spam every 5 minutes.
- If **RollingActivity.png** exists, it is attached to the alert email if **attach_plot = true**
- If **heartbeat.to** is blank, it uses **[email] to**.

Environment variable overrides are also available:

- **MAGNETOMETER_HEARTBEAT_ENABLED**
- **MAGNETOMETER_HEARTBEAT_HOUR_UTC**
- **MAGNETOMETER_HEARTBEAT_TO**
- **MAGNETOMETER_HEARTBEAT_ATTACH_PLOT**

Send a one-off heartbeat test email immediately

To verify heartbeat email delivery without waiting for **heartbeat.hour_utc**, run:

```
/bin/bash /home/pi/UKRAA_Magnetometer/scripts/testHeartbeatEmailACM0.sh
```

Or run the Python script directly:

```
/usr/bin/python3 /home/pi/UKRAA_Magnetometer/scripts/EvaluateAlertsACM0.py --test-heartbeat
```

This immediate heartbeat test does not update daily heartbeat schedule/state tracking.

Optional Remote FTP upload

Within the **/home/pi/UKRAA_Magnetometer/config** folder there is a file named **remote-upload.ini**.

By default the option of uploading plot PNG files is turned off (**false**).

Should the user wish to upload plot PNG files from their PicoMagnetometer to an externally hosted webpage follow the instructions below;

You can upload plot PNG files to an external hosting site while keeping local web publishing as the primary path.

Remote upload config file:

- **/home/pi/UKRAA_Magnetometer/config/remote-upload.ini**

Template created by installer:

- **install/remote-upload.ini.example**

Example settings:

```
[ftp]
enabled = true
site = your-uploader-fp
user = your-upload-user
password = your-upload-password
port = 21
directory = /data
timeout_seconds = 30
passive = true
create_dirs = true
upload_status_json = false
```

Behaviour:

- Daily run (09:30 via moveGraphs.sh) uploads:
  - **Activity.png**, **XYZ.png**, **HDZ.png**, **BI.png** to **/data**
- Rolling run (every 5 minutes via processRollingData.sh) uploads:
  - **RollingActivity.png**, **RollingXYZ.png, RollingHDZ.png**, **RollingBI.png** to **/data/rolling**
- Optional rolling status JSON upload:
  - set **upload_status_json = true**
  - uploads **data/status/current.json** to **/data/status/current.json**
  - external webpage status fallback (**status/current.json**) requires this to be true.
- Remote upload failures are non-blocking and do not stop local publishing.

Manual test commands:

/bin/bash /home/pi/UKRAA_Magnetometer/scripts/uploadRemoteACM0.sh daily
/bin/bash /home/pi/UKRAA_Magnetometer/scripts/uploadRemoteACM0.sh rolling

Combined test helper:

/bin/bash /home/pi/UKRAA_Magnetometer/scripts/testRemoteUploadACM0.sh

Updating the software

After a release has been published on GitHub, run the updater already installed on the RPi:

sudo bash ~/UKRAA_Magnetometer/scripts/updateMagnetometerACM0.sh

The updater downloads the latest GitHub release, checks that its tag matches the release **VERSION**, and updates the program files. It preserves recorded data, plot archives, configuration files, log files and temporary web files. It then runs **install.sh** to update the service, scheduled jobs and web files.

The updater requires an internet connection. It always selects the latest published GitHub release; it does not use ordinary repository checkout or **git pull**.

Publishing a release

For each release, update **VERSION** using the format **YYYY.MM.patch**, update **CHANGELOG.md**, commit the changes, and create a GitHub Release with a tag using exactly the same version, for example **2026.09.0**. The updater reads the latest published release and will stop if the tag and **VERSION** do not match.

The patch number starts at **0** for the first release in a month and increases for further releases in that month. Use a two-digit month, such as **2026.09.0**, rather than **2026.9.0**.

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Contact us

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